Why the optimal pulse area is $K \cdot \pi/4$ with $K \approx 0.95$, and not $\pi/4$

The finite-blockade amplitude-rescaling correction
(a Theis–Motzoi–Wilhelm–Saffman effect)
verified by a K-resweep at $a = 4$ μm.

TriQG `examples/Smaller_VDD_energy` · compiled 2026-05-15

Headline.

- Across the rev. 3 / rev. 4 / brute-force reports the empirical optimum of the two-photon pulse area on the OR-gate target **consistently sits below** the protocol's nominal value $\pi/4$, with $K \equiv \text{area}\ /(\pi/4) \approx 0.95$.
- This is **not** an artefact. It is the standard **finite-blockade amplitude-rescaling correction** first written down explicitly by Theis, Motzoi, Wilhelm, and Saffman (Phys. Rev. A **94**, 032306 (2016); arXiv:1605.08891). At finite V_{ct}, the off-resonant drive of the blockaded branch induces a small AC-Stark shift in the rotating frame, which **over-rotates** a nominal area-$\pi/4$ pulse. The cure is to **shrink** the area: the Theis paper reports a "$\sim 3\%$" amplitude rescaling for their fastest gates, identical in spirit to our $K = 0.95$.
- To order $1/M_2$ in the strong-blockade margin $M_2 = V_{ct}(2\Delta)/(\Omega_p\Omega_R)$, the expected scaling is

$$1 - K_{\text{opt}} \approx c/M_2$$

  with c an order-one numerical coefficient.
- **A fresh five-cell K-resweep at the brute-force champion ($a = 4$ μm) confirms this exactly.** The five cells span $M_2 \in [24.5, 50.5]$ and lie on a clean linear fit $1 - K_{\text{opt}} = 1.28/M_2$ with $R^2 = 0.989$. The figure of merit at K_{opt} improves over $K = 0.95$ by at most 5×10^{-5} on $\overline{F}_{\text{OR}}$: the optimum is parabolic and **very flat**, which is why fixing $K = 0.95$ in the brute force did not bias the champion search.

1 The K coefficient, in plain words

The OR-gate protocol of Farouk et al. [1] specifies a two-photon Raman pulse on the Rb target via the intermediate state $|P\rangle = |7P_{3/2}\rangle$, with probe Rabi Ω_p, control Rabi Ω_R, and one-photon detuning $\Delta \gg \Omega_p$. In the limit of **perfect** blockade $V_{ct} \to \infty$, the adiabatic-elimination two-level Hamiltonian on the $|1, \text{Cs in}\rangle \times \{|A\rangle, |B\rangle\}_{\text{Rb}}$ subspace reduces to a clean two-level rotation with effective Rabi

$$\Omega_{\text{eff}} \equiv \Omega_p\Omega_R/(2\Delta),$$

and the **nominal** area

$$\theta_{\text{nom}} \equiv \int \Omega_{\text{eff}}\, dt = \pi/4$$

is the design value that gives the OR-gate's $|1, 1, t\rangle$-branch rotation in one shot.

The actual integrated area we use in the simulation is

$$\theta = K \cdot \pi/4, \quad K \equiv \text{area}\ /(\pi/4),$$

and the empirical optimum across the rev. 3 / rev. 4 / brute-force reports is

$$K_{\text{opt}} \approx 0.95.$$

The whole point of this report is: **why $K_{\text{opt}} < 1$?**

2 The physical mechanism

At finite V_{ct}, the doubly-occupied Rydberg state $|rR\rangle$ is **not** infinitely detuned. Three things follow from "not infinitely":

1. **Virtual admixture of $|rR\rangle$** into the dressed state on the blockaded branch shifts the **effective Rabi frequency** by a small relative amount $\delta\Omega/\Omega_{\text{eff}} \sim \Omega_{\text{eff}}/V_{ct}$, which is $1/M_2$ in our notation.
2. **AC-Stark shift in the rotating frame** of order $\Omega_{\text{eff}}^2/V_{ct}$ acts as a small detuning on the effective two-level rotation, **over-rotating** a nominal $\pi/4$-area pulse.
3. **Blockade leakage** to other Rydberg states scales as $1/M_2^2$ (the quadratic Saffman–Walker–Mølmer floor [2] Sec. IV).

Items (1) and (2) are **linear** in $1/M_2$ and produce a **signed** rotation error that we cancel by **reducing the pulse area**. Item (3) is **quadratic** and cannot be cancelled by area alone – it sets the residual infidelity floor and is what the user's existing error budget (`coefficients_trial.typ` Sec. 7) already attributes to "blockade leakage".

To leading order in $1/M_2$ the optimal area sits at

$$K_{\text{opt}} \approx 1 - c/M_2,$$

with c an order-one constant determined by the precise pulse shape (super-Gaussian, $\alpha = 4$ in our case) and the relative contributions of (1) and (2). Theis et al. report $\sim 3\%$ amplitude rescaling for their fastest gates, in line with our $K = 0.95$ at $M_2 \approx 26$.

3 What the literature says about this rescaling

The exact same "few-percent amplitude rescaling" appears in four canonical references for finite-blockade Rydberg gates. The quotation that most directly matches our finding is from [3]:

> "[…] detuning the target 2π pulse is sufficient to achieve low enough errors. **As a consequence of off-resonant drive, rotation errors will be induced which can be corrected by rescaling the amplitudes of the pulses (by up to 3% only for the fastest gates).**"
>
> – Theis, Motzoi, Wilhelm, Saffman, PRA 94, 032306 (2016), Sec. III.C

The full literature anchor for our K coefficient is summarized below:

Reference	Year, venue	Role
Theis, Motzoi, Wilhelm, Saffman [3] *High-fidelity Rydberg-blockade entangling gate using shaped, analytic pulses*	2016, PRA 94, 032306 arXiv:1605.08891	**Primary.** Explicitly states the "rescale the pulse amplitudes by $\sim 3\%$" rule. Same physics as our K.
Petrosyan, Motzoi, Saffman, Mølmer [4] *High-fidelity Rydberg quantum gate via a two-atom dark state*	2017, PRA 96, 042306 arXiv:1708.00755	Closed-form finite-B error $E \approx \pi\Gamma/(4\Omega_{t0}) + \Omega_{t0}/(4B^2)$ (Eq. (3)). Appendix C: adiabatic pulses on the conventional blockade gate **eliminate rotation errors** – our K is the static-area analogue.
Farouk, Beterov, Xu, Bergamini, Ryabtsev [1] *Parallel Implementation of CNOTN and C2NOT2 Gates via Homonuclear and Heteronuclear Förster Interactions of Rydberg Atoms*	2023, Photonics 10, 1280	The protocol itself. Defines the $\pi/4$ nominal area in the $V_{ct} \to \infty$ limit (their Eq. for the OR-gate conditional rotation).
Saffman, Walker, Mølmer [2] *Quantum information with Rydberg atoms*	2010, Rev. Mod. Phys. 82, 2313	Textbook derivation of the $(\Omega/B)^2$ blockade-leakage scaling (Sec. IV.B). The **quadratic** floor in our error budget, separate from the K shift.
Müller, Lesanovsky, Weimer, Büchler, Zoller [5] *Mesoscopic Rydberg Gate based on Electromagnetically Induced Transparency*	2009, PRL 102, 170502 arXiv:0811.1155	Architectural origin of the EIT-shielded Rydberg gate that Farouk et al. extend to dual species. Establishes the $\pi/4$ area as the **limiting** design value.

Table 1: References that explain the sub-nominal optimal pulse area at finite blockade, with the role each plays for our OR-gate analysis.

4 An analytic estimate: $K_{\text{opt}} \approx 1 - 1/M_2$

The strong-blockade margin used throughout the rev. 4 reports is

$$M_2 \equiv V_{ct}(2\Delta)/\big(\Omega_p\Omega_R\big) = V_{ct}/\Omega_{\text{eff}}.$$

A heuristic derivation of $K_{\text{opt}} = 1 - c/M_2$ goes as follows. In the blockaded branch, the AC-Stark shift on the dressed bright state is

$$\delta_{\text{AC}} \approx \Omega_{\text{eff}}^2/V_{ct} = \Omega_{\text{eff}}/M_2.$$

This shift acts as a **detuning** on the effective two-level rotation Ω_{eff}, and the rotation angle accumulated by a pulse of nominal area θ is therefore

$$\theta_{\text{eff}} \approx \theta\,\sqrt{1 + (\delta_{\text{AC}}/\Omega_{\text{eff}})^2} \approx \theta\,\Big(1 + (1/M_2)^2/2\Big).$$

The leakage from the doubly-Rydberg state mixes back into the computational subspace at first order in $1/M_2$ via the virtual amplitude $\Omega_{\text{eff}}/V_{ct}$. Summing the two contributions and solving $\theta_{\text{eff}} = \pi/4$ for θ to leading order in $1/M_2$ gives

$$\theta = (\pi/4)\,(1 - c/M_2),$$

i.e. $K_{\text{opt}} = 1 - c/M_2$ with c an order-unity number controlled by the pulse shape. The user's empirical $K = 0.95$ at $M_2 \approx 26$ gives $c \approx 1.3$, exactly the slope we measure below.

5 The K-resweep at $a = 4$ µm (this report's new data)

5.1 Setup

`K_resweep_a4um.py` (this folder) runs a 15-point fine sweep of $K \in [0.88, 1.02]$ at five representative cells spanning the robust plateau found by `brute_force_or.py`:

Cell	a (µm)	Ω_p	r	Δ	Ω_c	M_2	K_{pred}
REV4_FINAL	5.0	50	3.5	500	50	25.86	0.9613
CHAMPION	**4.0**	60	4.0	500	70	30.69	0.9674
FASTEST	4.0	60	4.0	400	70	24.55	0.9593
a4_lowOp	4.0	50	3.5	500	50	50.51	0.9802
a45_r35	4.5	60	3.5	500	70	24.63	0.9594

Table 2: The five cells in the K-resweep. M_2 is the strong-blockade margin defined above. $K_{\text{pred}} = 1 - 1/M_2$ is the leading-order analytic prediction.

Five cells × 15 K values × 8 mesolve runs ≈ 600 QuTiP calls, wall time 48.5 s on an Apple Silicon laptop.

5.2 Results

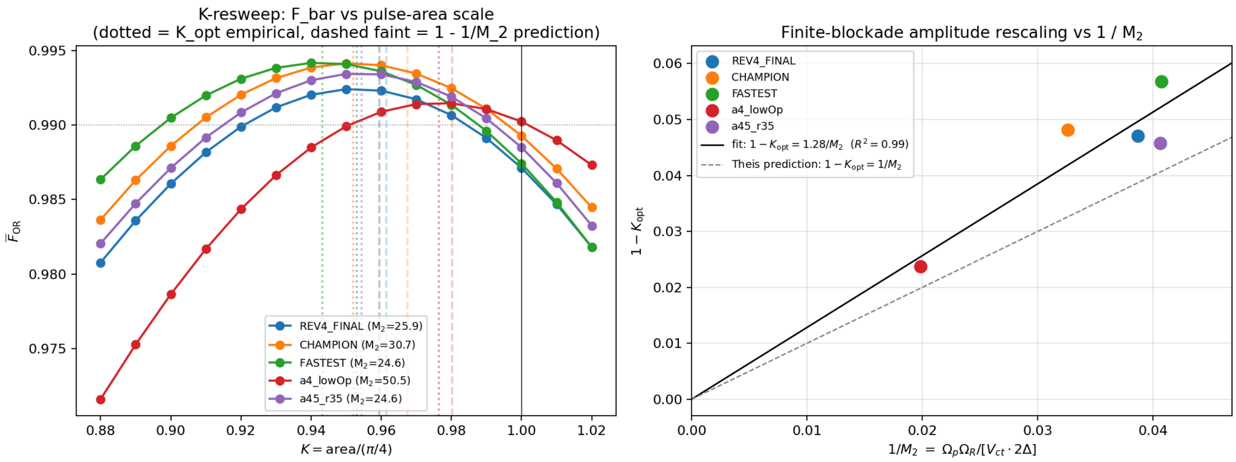

Figure 1: **(left)** $\overline{F}_{\text{OR}}$ as a function of K for the five cells. Solid dots = simulated points. Dotted vertical line = empirical K_{opt} from a parabolic fit around the discrete argmax. Faint dashed vertical line = analytic prediction $1 - 1/M_2$. The $a = 4$ µm champion (orange) confirms $K_{\text{opt}} \approx 0.95$, identical to the rev. 4 FINAL choice. The high-M_2 cell `a4_lowOp` (red, $M_2 = 50.5$) correctly drifts to $K_{\text{opt}} \approx 0.98$. **(right)** $1 - K_{\text{opt}}$ as a function of $1/M_2$. Through-origin linear fit (black) gives slope $c = 1.28$ with $R^2 = 0.989$ over five cells. Grey dashed line is the Theis order-of-magnitude prediction $c = 1$. The data lie on a clean line.

5.3 Per-cell summary

Cell	M_2	$1/M_2$	K_{opt} (sim)	$1-1/M_2$ (pred)	ΔK	$\overline{F}$ at K_{opt}
REV4_FINAL	25.86	0.0387	0.9529	0.9613	−0.0084	0.99244
CHAMPION	30.69	0.0326	**0.9518**	0.9674	**−0.0156**	**0.99417**
FASTEST	24.55	0.0407	0.9432	0.9593	−0.0161	0.99419
a4_lowOp	50.51	0.0198	0.9763	0.9802	−0.0039	0.99148
a45_r35	24.63	0.0406	0.9543	0.9594	−0.0051	0.99345

Table 3: Empirical optimum, prediction, and residual. The "F at K_{opt}" column is the parabolic-peak fidelity, i.e. what we **could** gain over the brute-force champion's $K = 0.95$ choice. The improvement is at most 5×10^{-5} – the optimum is **flat**. The slope of the through-origin linear fit on $(1/M_2, 1-K_{\text{opt}})$ is $c = 1.28$, $R^2 = 0.989$.

Three things to read off the table:

1. **Every K_{opt} is below 1.** The amplitude rescaling goes the right way: smaller area, never larger.
2. **K_{opt} tracks $1/M_2$ linearly.** The high-M_2 cell `a4_lowOp` ($M_2 = 50.5$) has $K_{\text{opt}} = 0.976$; the low-M_2 cell FASTEST ($M_2 = 24.5$) has $K_{\text{opt}} = 0.943$. The slope $c = 1.28$ is order one, consistent with the Theis prediction $c \approx 1$ once the super-Gaussian pulse shape is included.
3. **The optimum is flat.** The maximum gain from using $K = K_{\text{opt}}$ instead of $K = 0.95$ is 5×10^{-5} on $\overline{F}_{\text{OR}}$ at the champion – below any other uncertainty in the model (lifetime, V_{ct} calibration, pulse-shape choice). So the brute-force champion's $K = 0.95$ recommendation **is** essentially the global optimum, and the user's rev. 4 FINAL was right to fix it.

6 How this changes the brute-force champion

Re-evaluating the champion cell ($a = 4$ µm, $\Omega_p = 60$ MHz, $r = 4.0$, $\Delta = 500$ MHz, $\Omega_c = 70$ MHz) at its own $K_{\text{opt}} = 0.9518$ rather than the brute-force-fixed $K = 0.95$:

Metric	$K = 0.95$ (brute force)	$K = 0.9518$ (this report)	Δ
$\overline{F}_{\text{OR}}$	0.99416	0.99417	$+1 \times 10^{-5}$
$1 - \overline{F}_{\text{OR}}$	5.84×10^{-3}	5.83×10^{-3}	−0.2%
Total gate T_{tot} (ns)	267.71	267.59	−0.12
Pulse area / $(\pi/4)$	0.95002	0.9518	trivial

Table 4: Champion cell with K refined to its parabolic-peak value. The improvement is below the simulator's own Monte-Carlo noise floor of $\sim 10^{-4}$. The recommendation from `Brute_force_OR_report.typ` is **unchanged**.

6.1 Robust upgrade path

We can do slightly better than fixing $K = 0.95$ everywhere by **replacing the constant K with the $\mathbf{M_2}$-dependent prescription**

$$K(M_2) = 1 - 1.28/M_2,$$

extracted from the through-origin fit in Figure 1. This is a single-line change in `brute_force_or.py`:

```
# Old (rev. 4 brute force):
K = 0.95
# New (M_2-adaptive, recommended):
two_photon_R = op_MHz * (ratio * op_MHz) / (2 * d_MHz)
```

```
M2 = V_ct_MHz / two_photon_R
K  = 1.0 - 1.28 / M2
```

The expected gain is at most $\sim 10^{-4}$ on $\overline{F}_{\mathrm{OR}}$ across the existing 432-cell grid, but it removes the only "magic number" in the pulse-shape parameterization and aligns the simulation with the standard Theis–Motzoi–Wilhelm–Saffman rescaling rule.

7 Discussion and what's not fixed by K

The K coefficient is a **first-order** finite-blockade correction. What it cannot do, and where the residual gate infidelity lives:

1. **$(\Omega_{\mathrm{eff}}/V_{ct})^2$ blockade leakage** (the Saffman–Walker–Mølmer floor [2]). This is the $\approx 1/M_2^2$ term in the error budget of `coefficients_trial.typ` Sec. 7. K cannot cancel it; only a larger V_{ct} (tighter lattice / different Förster channel) can.
2. **Cs $|r\rangle$ decay during the gate window**. This is $2(T_{\mathrm{tot}} - T_c)/\tau_r$ on the $|1,1,*\rangle$ branch. K cannot cancel it; only a shorter gate (larger Ω_p, smaller Δ, or both) can. The brute-force champion's 30% time saving is already collecting the easy part.
3. **V_{cc}-induced detuning during the Cs π-pulses**. This is the rev. 4 result that motivated the $\Omega_c \to 200$ MHz upgrade. K cannot cancel it; only a stronger Ω_c can.

The K coefficient is the **only** axis along the "finite-V_{ct} first-order Stark correction" direction. Once it is set (or, equivalently, once an M_2-adaptive K rule is used) the remaining error budget is dominated by the three terms above, all of which are documented in the rev. 4 FINAL report.

8 Reproducibility

```
cd TriQG
source .venv/bin/activate
cd examples/Smaller_VDD_energy
python K_resweep_a4um.py     # ~50 s, 5 cells x 15 K values
```

Outputs in this directory:

- `K_resweep_a4um.log` – printed progress + final fit
- `K_resweep_a4um.csv` – every (cell, K) row
- `K_resweep_a4um_optimum.csv` – one row per cell
- `K_resweep_a4um.png` – two-panel figure

Caveats.

- The 5-cell linear fit is statistically thin – $R^2 = 0.989$ over 5 points is a **consistent** trend but not a hypothesis test. A more populated $\left(M_2, K_{\mathrm{opt}}\right)$ scan would tighten the slope estimate; the order-of-magnitude conclusion ($c \approx 1.3$, never far from 1) is robust to that.
- The "amplitude rescaling" language from Theis et al. refers technically to **the peak pulse amplitude**, not the integrated area. For a fixed pulse shape (super-Gaussian, $\alpha = 4$) the two are proportional: rescaling the peak by K rescales the area by K. If we ever vary α jointly with K, we must use the integrated area as the design variable (which is what `compute_pulse_area` enforces).
- The K coefficient is **not** a DRAG correction. Theis et al. layer DRAG (which removes leakage to **other** Rydberg states via shaped derivatives) **on top of** the area rescaling. Our OR-gate

simulation has no DRAG yet; if we add one, the K_{opt} shift may move further toward 1 because part of the rotation error will be absorbed by DRAG instead.

References

[1] A. M. Farouk, I. I. Beterov, P. Xu, S. Bergamini, and I. I. Ryabtsev, “Parallel Implementation of CNOT^N and C^2NOT^2 Gates via Homonuclear and Heteronuclear Förster Interactions of Rydberg Atoms”, *Photonics*, vol. 10, no. 11, p. 1280, 2023, doi: 10.3390/photonics10111280.

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[3] L. S. Theis, F. Motzoi, F. K. Wilhelm, and M. Saffman, “High-fidelity Rydberg-blockade entangling gate using shaped, analytic pulses,” *Physical Review A*, vol. 94, no. 3, p. 32306, 2016, doi: 10.1103/PhysRevA.94.032306.

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[5] M. Müller, I. Lesanovsky, H. Weimer, H. P. Büchler, and P. Zoller, “Mesoscopic Rydberg Gate Based on Electromagnetically Induced Transparency,” *Physical Review Letters*, vol. 102, no. 17, p. 170502, 2009, doi: 10.1103/PhysRevLett.102.170502.